M-estimation

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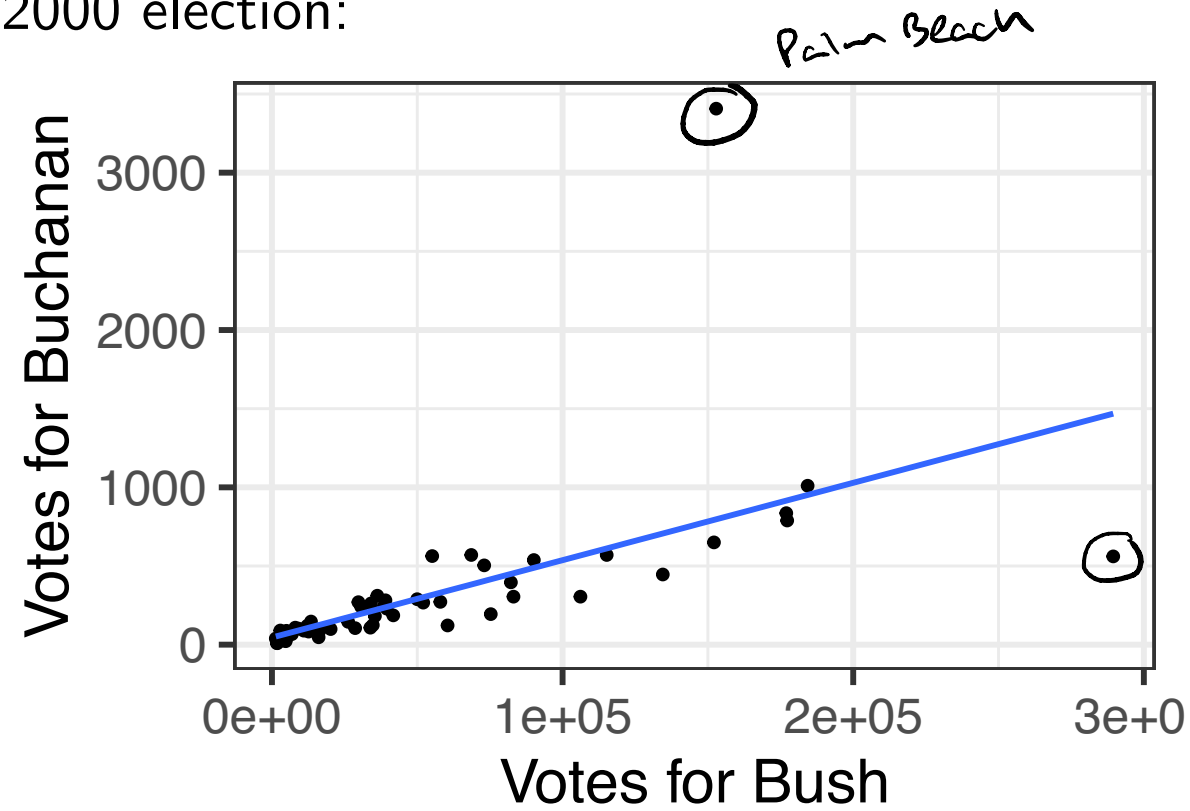
Motivating example

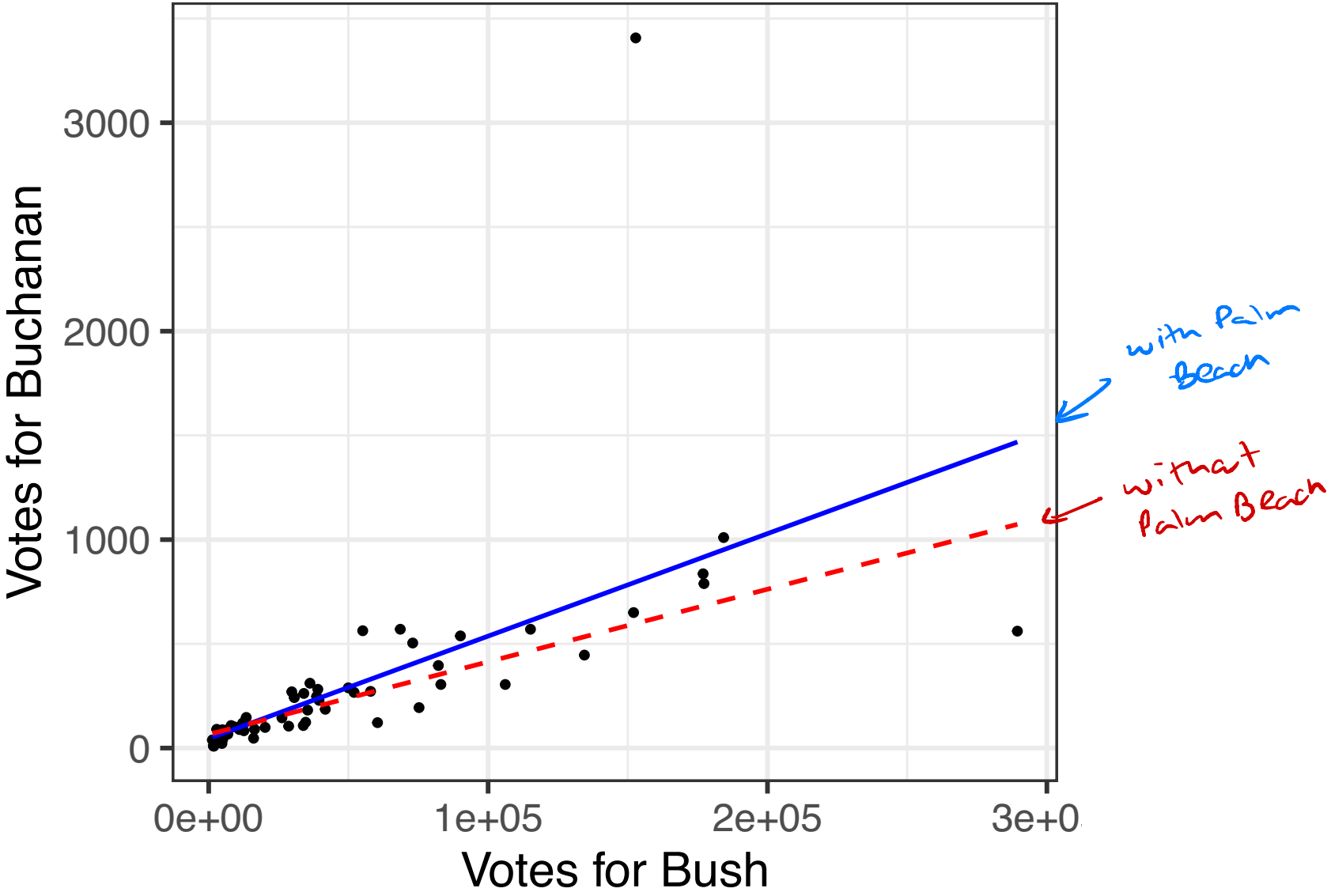
Votes for Pat Buchanan vs. votes for George Bush in Florida counties, 2000 election:



Question: What do you notice about this relationship? Anything concerning?

Motivating example

Comparing model with and without Palm Beach:



Robust linear regression

Model: $Y_i = \mathbf{x}_i^T \beta + \varepsilon_i$

OLS: $\widehat{\beta}$ minimizes

$$\sum_{i=1}^{n} (Y_i - \mathbf{x}_i^T \beta)^2$$

squaring makes big residuals really big (outliers have a big influence)

Robust regression: $\widehat{\beta}$ minimizes

$$\sum_{i=1}^{n} \rho(Y_i - \mathbf{x}_i^T \beta)$$

still assume $Y_i = x_i^T \beta + \varepsilon_i$ but use a different criterion to choose $\hat{\beta}$

One choice of ρ is the **Huber loss function**:

$$\rho_K(u) = \begin{cases} \frac{1}{2} u^2 & |u| \le K \\ K|u| - \frac{1}{2} K^2 & \text{else} \end{cases}$$

less emphasis on big residuals

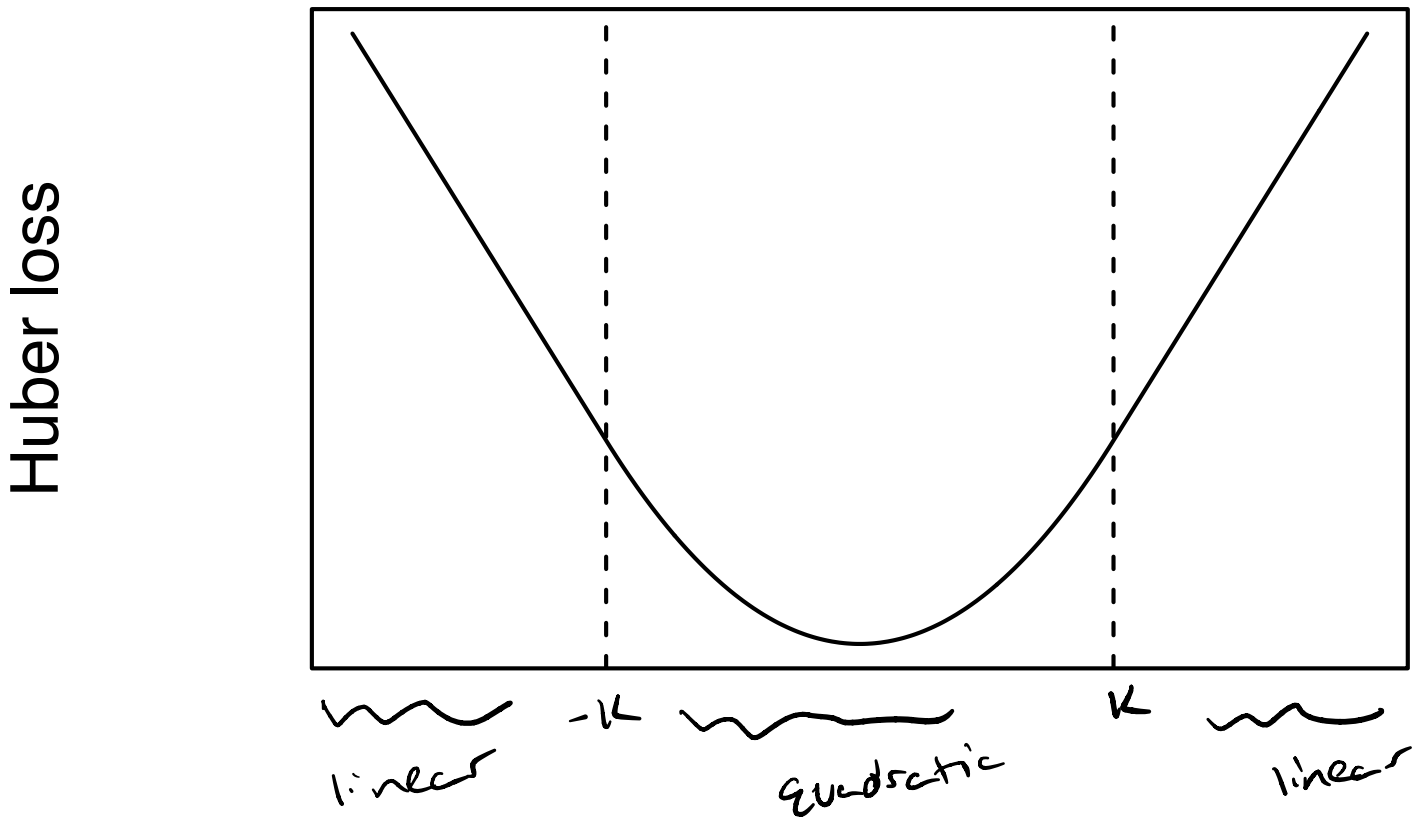
as $K \to \infty$, becomes OLS

as $K \to 0$, becomes L_1 regression (minimize $\sum_i |Y_i - x_i^T \beta|$)

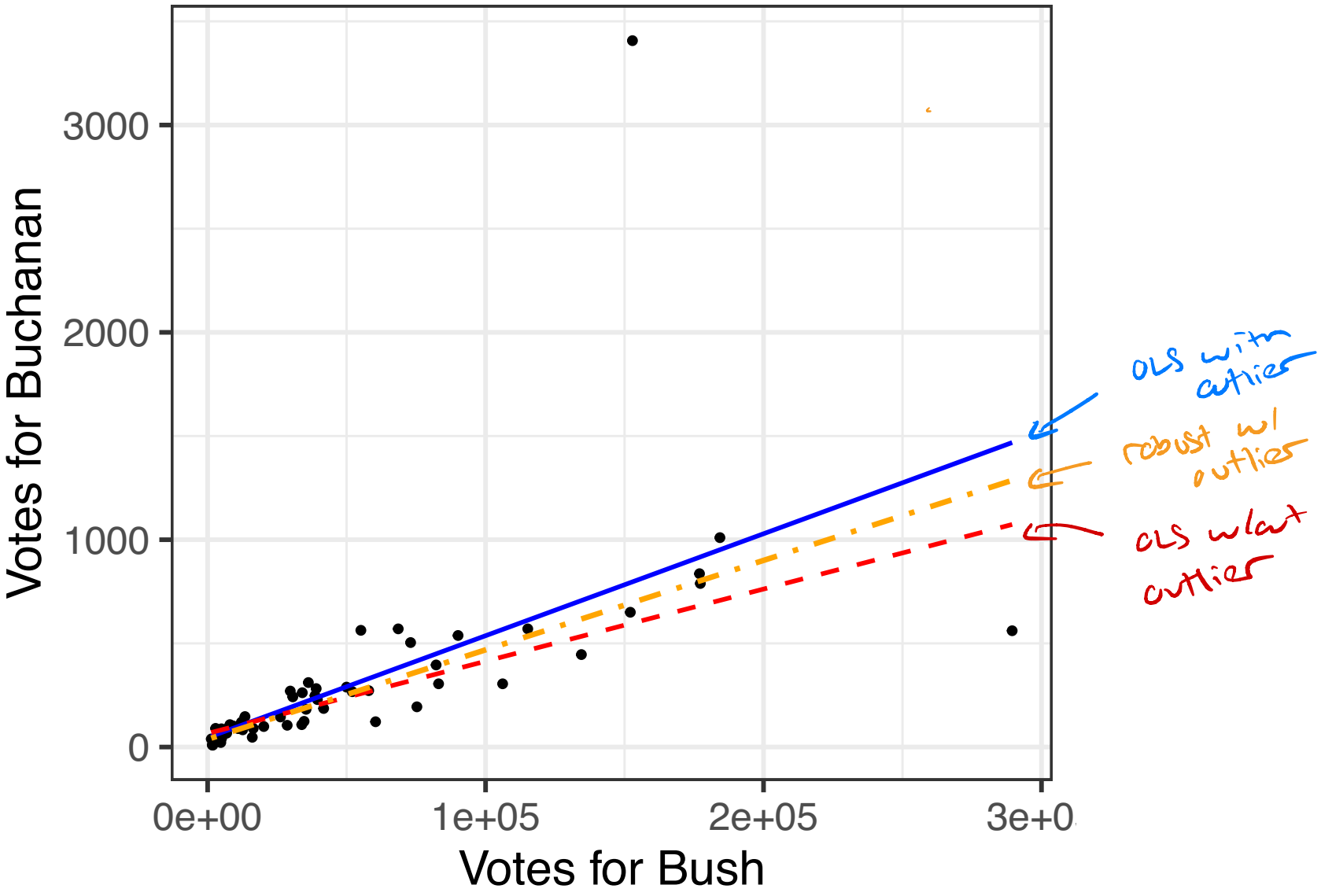
Huber loss function

Standard choice for k: $1.345 \times \hat{\sigma}$
(=> ARE for robust vs. mean if data are normal is 0.96)

$$\rho_k(u) = \begin{cases} \frac{1}{2}u^2 & |u| \le k \\ k|u| - \frac{1}{2}k^2 & \text{else} \end{cases}$$



Robust linear regression



Robust linear regression

$$|u| = \begin{cases} -u & u<0 \\ u & u>0 \end{cases}$$

$$\frac{d}{du}|u| = \begin{cases} -1 & u<0 \\ 1 & u>0 \end{cases} = \text{sign}(u)$$

Model: $Y_i = \mathbf{x}_i^T \beta + \varepsilon_i$

$\widehat{\beta}$ minimizes $\sum_{i=1}^{n} \rho_k(Y_i - \mathbf{x}_i^T \beta)$

$$\frac{\partial}{\partial \beta} \sum_{i=1}^{n} \rho_k(Y_i - x_i^T \beta) \overset{\text{set}}{=} 0$$

$$\Rightarrow \sum_{i=1}^{n} -\rho_k'(Y_i - x_i^T \beta)\, x_i = 0$$

$$\Rightarrow \hat{\beta} \quad \text{solve} \quad \sum_{i=1}^{n} \rho_k'(Y_i - x_i^T \beta)\, x_i = 0$$

$$\rho_k(u) = \begin{cases} \frac{1}{2}u^2 & |u| \le k \\ k|u| - \frac{1}{2}k^2 & \text{else} \end{cases} \Rightarrow \rho_k'(u) = \begin{cases} u & |u| \le k \\ k\,\text{sign}(u) & \text{else} \end{cases}$$

(use IRLS to estimate $\hat{\beta}$)

M-estimation

"maximum-likelihood type"

So far, a lot of our estimators look something like this: $\widehat{\theta}$ solves

$$\sum_{i=1}^{n} \psi(Y_i, \theta) = \mathbf{0}$$

Sample mean: $\hat{\mu}$ solves $\sum_{i=1}^{n} (Y_i - \mu) = 0$

MLE: $\hat{\theta}$ solves $\sum_{i=1}^{n} \frac{\partial}{\partial \theta} \log f(Y_i | \theta) = 0$

Linear regression: (OLS) $\hat{\beta}$ solves $\sum_{i=1}^{n} (Y_i - x_i^T \beta) x_i = 0$

Logistic regression: $\hat{\beta}$ solves $\sum_{i=1}^{n} (Y_i - p_i) x_i = 0$

Robust regression: $\hat{\beta}$ solves $\sum_{i=1}^{n} \rho'(Y_i - x_i^T \beta) x_i = 0$

Can we say something **in general** about estimators of this type?

M-estimation

Let $Y_1, \ldots, Y_n$ be iid samples from some distribution g. (Y_i could be a scalar or vector random variable). Let $\psi(Y, \boldsymbol{\theta})$ be some suitably smooth function, which depends on a parameter $\boldsymbol{\theta} \in \mathbb{R}^d$.

Let $\hat{\theta}$ solve $\sum_{i=1}^{n} \psi(Y_i, \theta) = 0$

Let θ_0 solve $\mathbb{E}_g[\psi(Y_i, \theta)] = 0 \Rightarrow \mathbb{E}_g[\psi(Y_i, \theta_0)] = 0$

Under regularity conditions:

1. $\hat{\theta} \xrightarrow{p} \theta_0$
2. $\sqrt{n}(\hat{\theta} - \theta_0) \xrightarrow{d} N(0, S(\theta_0))$

$S(\theta_0) = A^{-1} B A^{-1}$

$A = -\mathbb{E}_g[\psi'(Y_i, \theta_0)]$ $\quad B = \mathbb{E}_g[\psi(Y_i, \theta_0)\psi(Y_i, \theta_0)^T]$

$\hat{A} = -\frac{1}{n} \sum_{i=1}^{n} \psi'(Y_i, \hat{\theta})$ $\quad \hat{B} = \frac{1}{n} \sum_{i=1}^{n} \psi(Y_i, \hat{\theta})\psi(Y_i, \hat{\theta})^T$

Example: robust linear regression

$$\psi(Y_i, \mathbf{x}_i, \beta) = \rho_k'(Y_i - \mathbf{x}_i^T \beta)\mathbf{x}_i$$

$$\rho_k(u) = \begin{cases} \frac{1}{2}u^2 & |u| \le k \\ k|u| - \frac{1}{2}k^2 & \text{else} \end{cases} \qquad \rho_k'(u) = \begin{cases} u & |u| \le k \\ k\ \text{sign}(u) & \text{else} \end{cases}$$

β^0 solves $\mathbb{E}_g[\rho_k'(Y_i - x_i^T\beta)x_i] = 0$

$$A = -\mathbb{E}_g\left[\frac{\partial}{\partial \beta^0}\{\rho_k'(Y_i - x_i^T\beta^0)x_i\}\right]$$

$$= \mathbb{E}_g[\rho_k''(Y_i - x_i^T\beta^0)x_i x_i^T] \qquad \rho_k''(u) = \begin{cases} 1 & |u| \le k \\ 0 & \text{else} \end{cases}$$

$$= \mathbb{1}\{|u| \le k\}$$

$$B = \mathbb{E}_g[(\rho_k'(Y_i - x_i^T\beta^0))^2 x_i x_i^T]$$